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Specification

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1 PURPOSE AND GUIDANCE

This document is made to outline the discussions with **Attalon** on the CRIMs (gaps) carried out related to IFS® functionality. In addition, this document lays out the processes **Attalon** practice currently and possible changes to the workflows and possible solutions under the IFS®.

This document specifies:

- Attalon business requirements
- How screens, views, fields, and tables shall look and operate.
- How business logic and functions shall operate and/or calculate
- How the application shall be customized to solve the business requirements.

The **Solution Description** defines **what** will be delivered. The solution is matched against the requirements described in the Requirements section.

The **Delimitations** describe what is explicitly **not covered** by this CRIM. Together with the Solution description, these sections delimit the extent of the functional addition agreed for implementation.

2 OPEN ISSUES/DECISIONS REQUIRED

This specification cannot be completed until the following issues/decisions have been resolved:

Issue/Decision	Responsible

3 BUSINESS REQUIREMENTS DETAILS

3.1 BUSINESS REQUIREMENTS

- Attalon requires that IFS Freight Interface functionality, APIs, Basic Data Tables and interface to the larger IFS ERP applications be leveraged as much as possible for this specification. The primary development effort is leveraging the "Freight Provider" functionality in IFS for purpose of interfacing with FedEx for quoting freight cost and price.
 - Existing Freight Interface in IFS includes:
 - Shipment Management / Freight Integration
 - Basic Data
 - Freight Provider (i.e., use provider Pacejet to understand how to setup provider FedEx)
 - In separate specification (I-007) replace with UPS
 - Shipment Services
 - Examples are overnight, next day, etc.
 - Freight Carriers
 - Examples are: FedEx, UPS, FedEx Over Night, ect.
 - Freight Delivery Terms
 - Examples are pre-paid and add (PPAD)
 - Freight Transactions
 - All Financial transactions for the freight interface.

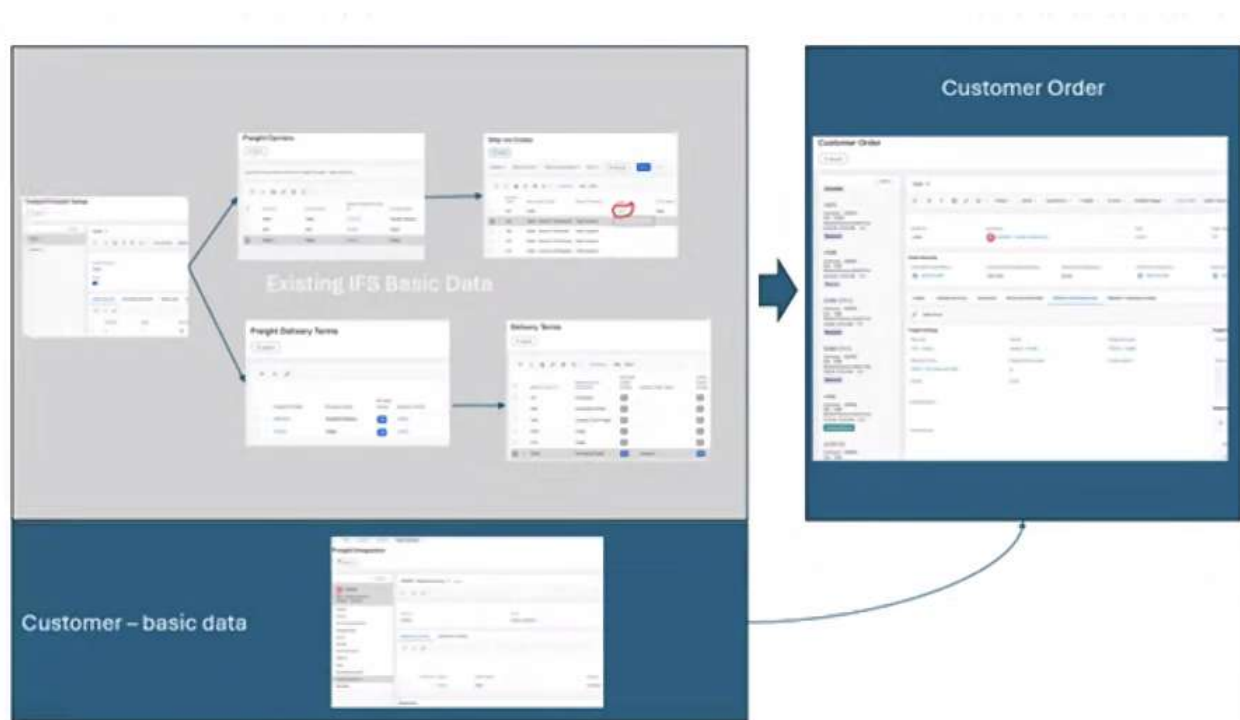
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- May include calls / return payloads from FedEx or any provider
- Freight Log
 - All calls / return payloads from FedEx or any provider
- Attalon requires that at Shipment (Shipment Header Screen), the shipping department will be able to automate a call to either FedEx or UPS and receive the return values for cost and price for a specific shipment and service.
- Attalon requires that the shipment cost and price be able to be marked up (%)
- Attalon requires that the 'selected' shipment service (FedEx, UPS) be able to be selected and then inserted into the original customer order charge line and linked to the customer order line found on the first shipment line of the shipment.
 - The COL charges be updated or inserted with the cost and markup price.
- Attalon requires that an integration to FedEx (and UPS) be able to call FedEx (and UPS) open interfaces (RESTful API) and retrieve information regarding shipment.
 - Interface (POST) will send to FedEx (and UPS)
 - Deliver From Address
 - Deliver To Address
 - Service Type (i.e. overnight, same-day, ground, etc.)
 - Handling Unit/s for Shipment (Pallets, Boxes, etc.)
 - Type Handling Unit/s (Pallet, Box)
 - Size Handling Unit/s (L, W, H)
 - Weight of Handling Unit/s
 - Shipment Date
 - Special services: overnight, ...
 - Interface will receive an initial estimate for booking to the Shipment Freight Transactions Line
 - Shipment Reviews Estimate and then once estimate is selected, it will book the FedEx shipment assigning the correct charge to the correct charge line associated with the correct customer order line.
- Attalon requires that returned FedEx 'estimate' be able to be selected and booked.
 - The Customer Order Charge Line is created which is linked to the First Customer Order Line found on the shipment.
 - The charge cost is the cost of freight in return payload from FedEx cost
 - The charge price will be the charge cost x markup found on the Freight Delivery Terms.
 - The sales charge used on the CO charge line will be found in the Freight Delivery Terms.
 - The Shipment Header is updated
 - Planned Ship Date is updated with Pickup date, time
 - PRO No is updated with Tracking Number (master or first tracking No)
- Attalon requires that a background job run once per day to transmit to the customer that the shipment has occurred. It will only transmit those shipments, shipped (i.e., delivered) that day to the customer.

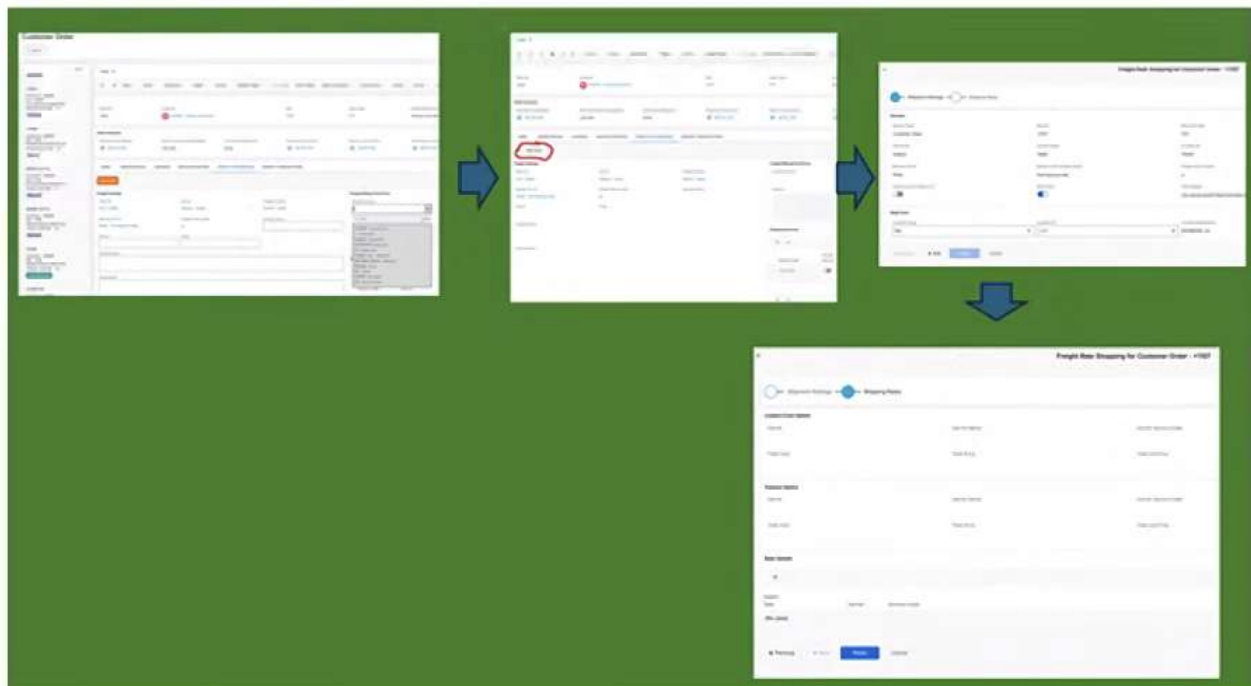
4 SOLUTION DESCRIPTION

4.1 IFS FREIGHT INTERFACE BASIC DATA SETUP



Specification

4.2 PROCESS FLOW FOR INTERFACE



4.4 PROCESS FLOW BUSINESS LOGIC

- Initial Call to FedEx
 - Organize Payload based on L, W, H of boxes (or pallets)
 - Total Qty of boxes (or pallets) by L, W, H
 - Total Weight of boxes (or pallets) by L, W, H
 - Ship from address (on shipment header), ship to address (on shipment header), and service
 - Shipment Date
 - Call FedEx -> Send Payload
 - Receive Return Payload
 - Insert Return Payload into the Shipment / Tab "Freight Transactions" Screen.
 - If return is an array of potential pickup dates, times, and/or services then insert into multiple lines in table.
- Select and Book
 - (Note, that this may require a secondary confirmation/booking call to FedEx)
 - Using delivery terms on the shipment, lookup freight delivery terms. Using the freight delivery terms lookup the sales charge linked to the freight delivery terms.
 - Using the sales charge and the source ref 1, 2, 3, 4 of the first shipment line, insert the sales charge on the customer order linked to the customer order line.
 - Using the cost in the return payload, calculate the price by using markup x price. (i.e. \$100 x 1.5 = \$150)
 - Insert or update the charge line cost and price
 - Update the shipment header / General / Reference Documents and Misc. Info/ Field PRO No with the tracking number.
 - Update the shipment header / Planned Ship Date Time with the estimated pickup date and time in return payload.

4.5 FEDEX DEV

Obtaining a FedEx API Key via the FedEx Developer Portal

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You can get a FedEx API Key (Client ID) and Secret Key (Client Secret) by creating a project in the FedEx Developer Portal. Test credentials are provided first, and production keys require additional steps.

Getting Test API Credentials

Go to the [FedEx Developer Portal](#) and log in with your account, or sign up if you don't have one.

In the left-hand menu, click **My Projects**.

Select **Create New Project**.

Choose the reason for the API from the dropdown and enter a unique project name.

Select the APIs you want (e.g., Ship API, Rates and Transit Times, Address Validation).

Enter project details, including name and shipping location.

Select the countries you plan to ship within so FedEx can assign a test shipping account.

Confirm the details to complete project creation.

View your **API Key (Client ID)** and **Secret Key (Client Secret)** in the confirmation window (these are test credentials).

Generating Production API Credentials

In the FedEx Developer Portal, click **My Projects**.

Open the project you created for test credentials.

Go to the **Production Key** tab.

Select your account number and click **Next**.

If the account is not added, click **Add shipping account** and provide account number, nickname, and billing address.

Once ready, access your production API Key and Secret Key in the **Production** tab.

Important Notes

FedEx Certification is mandatory for Ship API and Open Ship API before production use.

For authentication, use the OAuth endpoint with your Client ID and Client Secret to generate an access token.

Test Mode must be enabled when using sandbox credentials to avoid generating live labels.

4.6 EXAMPLE OF PAYLOAD TO FEDEX

Message Sent --> IFS to FedEx

```
{"accountNumber":{"value":"108993660"},"rateRequestControlParameters":{"returnTransitTimes":"true","servicesNeededOnRateFailure":"false","rateSortOrder":"SERVICENAMETRADDITIONAL"},"requestedShipment":{"shipper":{"address":{"streetLines":["4704 Interstate Blvd"],"city":"Loves Park","stateOrProvinceCode":"IL","postalCode":"62111","countryCode":"US","residential":"false"},"recipient":{"address":{"streetLines":["2340 S. Crockett Parkway"],"city":"JIMSCITY","stateOrProvinceCode":"TN","postalCode":"39813","countryCode":"US","residential":"false"},"rateRequestType":["ACCOUNT"],"shipDateStamp":"2025-11-11","pickupType":"USE_SCHEDULED_PICKUP","requestedPackageLineItems":[{"weight":{"units":"LB","value":50},"dimensions":{"length":12,"width":11,"height":11,"units":"IN"}},{"weight":{"units":"LB","value":11},"dimensions":{"length":14,"width":8,"height":6,"units":"IN"}}],"packagingType":"YOUR_PACKAGING","totalPackageCount":2}}
```

5 SCREEN SHOT EXAMPLES

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Home > Shipping > Shipment Delivery > Shipment 14798

Shipment

Search Advanced Search with Time Zone

Status Shipment ID Notes Receiver ID Sender Address ID More Favorites Search Saved Searches

Sort by

Receiver ID: 100299
Receiver Description: AutoZone

194838
Cancelled
Receiver ID: 104501
Receiver Description: VulcanGroup, Inc.

4482
Cancelled
Receiver ID: LP-INTERNAL-CUST
Receiver Description: Loves Park Internal Customer

3648
Cancelled
Receiver ID: LP-INTERNAL-CUST
Receiver Description: Loves Park Internal Customer

2657
Cancelled
Receiver ID: 100299
Receiver Description: AutoZone

14798
Closed
Receiver ID: ROC-INTERNAL-CUST
Receiver Description: Rockford Internal Customer

Sender

Sender Type	Sender ID	Sender Description
Site	OH	Des Plaines

Receiver

Receiver Type	Receiver ID	Receiver Description
Customer	ROC-INTERNAL-CUST	Rockford Internal Customer

Settings and Indicators

SHIPMENT LINES AVAILABLE LINES ADDRESS GENERAL FREIGHT INTEGRATION FREIGHT TRANSACTIONS

Rate Shop

Freight Settings

Ship Via	Carrier	Freight Provider
217 - OH ROBOCOIN - GROUND	OH ROBOCOIN - OH Roberson	AGT RALD - Unrated Carrier

Freight Billing/Third Party

Freight Account

Address

Shipping Services

Service Code	Include Service
(No data)	

Home > Shipping > Shipment Delivery > Shipment 14798

Shipment

Search Advanced Search with Time Zone

Status Shipment ID Notes Receiver ID Sender Address ID More Favorites Search Saved Searches

Sort by

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Receiver ID: LP-INTERNAL-CUST
Receiver Description: Loves Park Internal Customer

2657
Cancelled
Receiver ID: 100299
Receiver Description: AutoZone

14798
Closed
Receiver ID: ROC-INTERNAL-CUST
Receiver Description: Rockford Internal Customer

Shipment ID: 14798
Site: OH
Source Ref Type: Customer Order
Next Step in Shipment Flow

Shippers Type: null
Created: 2/10/2026
Planned Ship Date/Time: 2/10/26, 12:00 AM CST
Planned Delivery Date/Time

Notes: Dispatch Advice History

Sender

Sender Type	Sender ID	Sender Description
Site	OH	Des Plaines

Receiver

Receiver Type	Receiver ID	Receiver Description
Customer	ROC-INTERNAL-CUST	Rockford Internal Customer

Settings and Indicators

SHIPMENT LINES AVAILABLE LINES ADDRESS GENERAL FREIGHT INTEGRATION FREIGHT TRANSACTIONS

Freights Transactions

Send to NEXT RALD

State	Error	Freight Message Type	Freight Provider ID	Delivery Link	Create Date	Total Weight	Total Price	Total Cost	Total U.S. Price	Currency Code
(No data)										

Specification

[illegible][illegible][illegible]

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Specification

Shipment Management > Freight Integration > Basic Data > Freight Provider Setup

Freight Provider Setup

Q Search

sort by

FREIGHTWISE

FEDEX

NOT RATED

UPS

SCALE

PACEJET

FEDEX

Freight Provider ID: FEDEX

Name: FedEx Web Services

Ship Contact Role: [dropdown]

Ship Contact Role Name: [dropdown]

Active: ☒

Default Provider: No

CONFIG VALUES PROVIDER LOCATIONS TERMS CODE MESSAGE MAPPING

Config ID	Value	Description
ACCOUNT_NUMBER	108923660	
MAX_CARTON_CAPACITY	10	
MAX_WEIGHT_CAPACITY	200	

Shipment Management > Freight Integration > Basic Data > Freight Provider Setup

Freight Provider Setup

Q Search

sort by

FREIGHTWISE

FEDEX

NOT RATED

UPS

SCALE

PACEJET

FEDEX

Freight Provider ID: FEDEX

Name: FedEx Web Services

Ship Contact Role: [dropdown]

Ship Contact Role Name: [dropdown]

Active: ☒

Default Provider: No

CONFIG VALUES PROVIDER LOCATIONS TERMS CODE MESSAGE MAPPING

Site	Sender Type	Sender ID	Sender Description	Freight Size Code	Freight Facility	Note
(No data)						